COMP1661 Logbook Upload Template

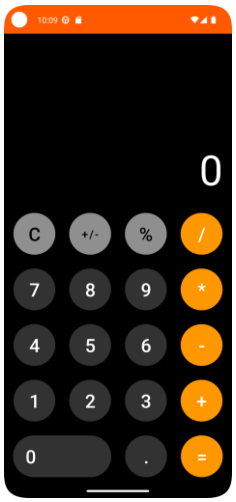
1. Basic Information

1.1 Student name	**Marzuq Sogbanmu**
1.2 Who did you work with? Note that for logbook exercises you are allowed to work with one other person as long as you give their name and login id and both contribute to the work.	**Name: N/A** **Login id: N/A**
1.3 Which Exercise is this? Tick as appropriate.	• Exercise 1 ☒ • Exercise 2 ☐ • Exercise 3 ☐
1.4 How well did you complete the exercise? Tick as appropriate.	• I tried but couldn't complete it ☐ • I did it but I feel I should have done better ☐ • I did everything that was asked ☒ • I did more than was asked for ☐
1.5 Briefly explain your answer to question 1.4	**Everything that was asked of the exercise I was able to achieve. However, I wanted to implement additional features (plus/minus button, percentage button) but was unable to complete the functionality for them.**

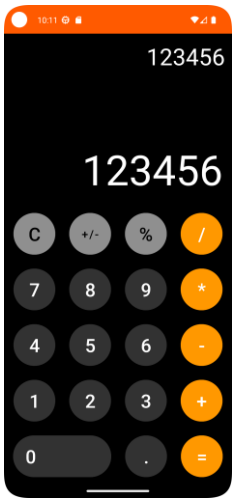
2. Exercise answer

2.1 Screen shots demonstrating what you achieved

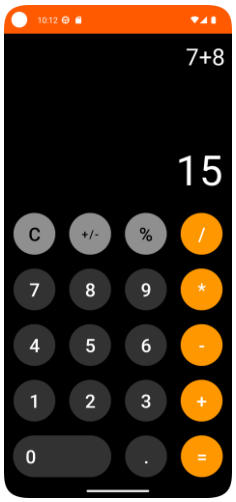
Default layout



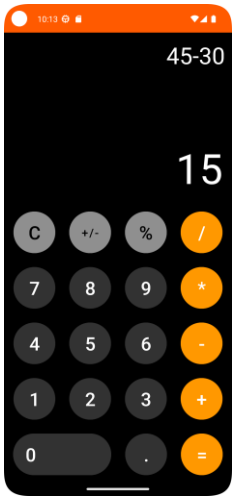
Displaying of numbers pressed



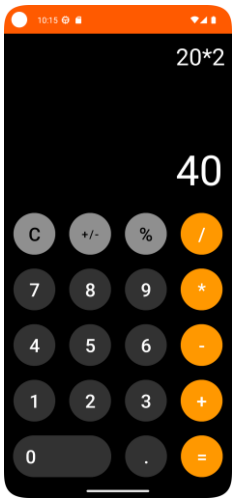
Addition



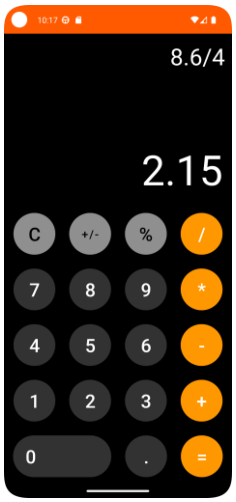
Subtraction



Multiplication



Division



2.2 Code that you wrote

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/black"
    tools:context=".MainActivity">

    <!--Definition of method text-->
    <TextView
        android:id="@+id/txt_method"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_above="@id/txt_result"
        android:layout_margin="12dp"
        android:layout_marginTop="12dp"
        android:layout_marginBottom="0dp"
        android:text=""
        android:textAlignment="textEnd"
        android:textColor="@color/white"
        android:textSize="40dp" />

    <!--Definition of results text-->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/txt_result"
        android:text="0"
        android:textSize="72dp"
        android:textAlignment="textEnd"
        android:textColor="@color/white"
        android:layout_above="@id/buttons_layout"
        android:layout_margin="16dp"/>

<!--Declaration of button layout-->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:background="@color/black"
        android:layout_alignParentBottom="true"
        android:id="@+id/buttons_layout">

<!--Definition of individual rows-->
        <!--Row 1-->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center"
            android:orientation="horizontal">

            <!--Definition of button attributes-->
            <com.google.android.material.button.MaterialButton
                android:id="@+id/btn_clear"
                style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
                android:layout_width="72dp"
                android:layout_height="72dp"
                android:layout_margin="12dp"
                android:backgroundTint="#8F8F8F"
                android:text="C"
                android:textColor="@color/black"
```

```
            android:textSize="32dp"
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_plusminus"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#8f8f8f"
            android:text="+/-"
            android:textColor="@color/black"
            android:textSize="20sp"
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_percent"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#8f8f8f"
            android:text="%"
            android:textColor="@color/black"
            android:textSize="32dp"
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_divide"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#FF9800"
            android:text="/"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />

    </LinearLayout>

    <!--Row 2-->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:orientation="horizontal">

        <!--Definition of button attributes-->
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_7"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#323232"
            android:text="7"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_8"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#323232"
            android:text="8"
            android:textColor="@color/white"
            android:textSize="32dp"
```

```
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_9"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#323232"
            android:text="9"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_multiply"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#FF9800"
            android:text="*"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />

    </LinearLayout>

    <!--Row 3-->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:orientation="horizontal">

        <!--Definition of button attributes-->
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_4"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#323232"
            android:text="4"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_5"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#323232"
            android:text="5"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />

        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_6"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#323232"
            android:text="6"
            android:textColor="@color/white"
            android:textSize="32dp"
```

```
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_subtract"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#FF9800"
            android:text="-"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />

    </LinearLayout>

    <!--Row 4-->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:orientation="horizontal">

        <!--Definition of button attributes-->
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_1"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#323232"
            android:text="1"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_2"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#323232"
            android:text="2"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_3"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#323232"
            android:text="3"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btn_addition"
            style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:layout_margin="12dp"
            android:backgroundTint="#FF9800"
            android:text="+"
            android:textColor="@color/white"
            android:textSize="32dp"
            app:cornerRadius="36dp" />
```

```
        </LinearLayout>

        <!--Row 5-->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center"
            android:orientation="horizontal">

            <!--Definition of button attributes-->
            <com.google.android.material.button.MaterialButton
                android:id="@+id/btn_0"
                style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
                android:layout_width="169dp"
                android:layout_height="72dp"
                android:layout_margin="12dp"
                android:backgroundTint="#323232"
                android:text="0"
                android:textAlignment="textStart"
                android:textColor="@color/white"
                android:textSize="32dp"
                app:cornerRadius="36dp" />
            <com.google.android.material.button.MaterialButton
                android:id="@+id/btn_decimal"
                style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
                android:layout_width="72dp"
                android:layout_height="72dp"
                android:layout_margin="12dp"
                android:backgroundTint="#323232"
                android:text="."
                android:textAlignment="center"
                android:textColor="@color/white"
                android:textSize="32dp"
                app:cornerRadius="36dp" />
            <com.google.android.material.button.MaterialButton
                android:id="@+id/btn_equals"
                style="@style/Widget.MaterialComponents.ExtendedFloatingActionButton"
                android:layout_width="72dp"
                android:layout_height="72dp"
                android:layout_margin="12dp"
                android:backgroundTint="#FF9800"
                android:text="="
                android:textColor="@color/white"
                android:textSize="32dp"
                app:cornerRadius="36dp" />

        </LinearLayout>

    </LinearLayout>

</RelativeLayout>
```

MainActivity.java

```
package com.example.a001102232cw2_calculator;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.TextView;

import com.google.android.material.button.MaterialButton;
import org.mozilla.javascript.Context; //From Mozilla's Rhino engine
import org.mozilla.javascript.Scriptable; //From Mozilla's Rhino engine

public class MainActivity extends AppCompatActivity implements View.OnClickListener {

    TextView resultText, methodText;
    MaterialButton clearButton, plusminusButton, percentButton;
    MaterialButton divideButton, multiplyButton, subtractButton, addButton,
equalsButton;
    MaterialButton button0, button1, button2, button3, button4, button5, button6,
button7,button8, button9;
    MaterialButton  decimalButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

//      Getting the IDs for the textviews and buttons
        resultText = findViewById(R.id.txt_result);
        methodText = findViewById(R.id.txt_method);

        assignID(clearButton, R.id.btn_clear);
        assignID(plusminusButton, R.id.btn_plusminus);
        assignID(percentButton, R.id.btn_percent);

        assignID(divideButton, R.id.btn_divide);
        assignID(multiplyButton, R.id.btn_multiply);
        assignID(subtractButton, R.id.btn_subtract);
        assignID(addButton, R.id.btn_addition);
        assignID(equalsButton, R.id.btn_equals);

        assignID(button0, R.id.btn_0);
        assignID(button1, R.id.btn_1);
        assignID(button2, R.id.btn_2);
        assignID(button3, R.id.btn_3);
        assignID(button4, R.id.btn_4);
        assignID(button5, R.id.btn_5);
        assignID(button6, R.id.btn_6);
        assignID(button7, R.id.btn_7);
        assignID(button8, R.id.btn_8);
        assignID(button9, R.id.btn_9);

        assignID(decimalButton, R.id.btn_decimal);
    }

    void assignID(MaterialButton btn, int id) {
        btn = findViewById(id);
        btn.setOnClickListener(this);
    }

    @Override
    public void onClick(View view) {
//      Displaying the text from buttons to the method textview
        MaterialButton button = (MaterialButton) view;
```

```
        String buttonText = button.getText().toString();
        String calculation = methodText.getText().toString();


// Button functionality
        // Clear button
        if(buttonText.equals("C")){
            methodText.setText("");
            resultText.setText("0");
            return;
        }
//      Plus/minus button (unfinished)
        if (buttonText.equals("+/-") && !calculation.startsWith("-")) {
            methodText.append("-");
            methodText.setText(calculation);
        } else if (buttonText.equals("+/-") && calculation.startsWith("-")) {
            calculation = calculation.substring(1);
        }

//      Equals button
        if (buttonText.equals("=")){
            methodText.setText(resultText.getText());
            return;
        } else {
            calculation = calculation + buttonText;
        }
        methodText.setText(calculation);

        String final_result = getResult(calculation);

        if (!final_result.equals("Calculation error")) {
            resultText.setText(final_result);
        }
    }

    String getResult (String data) {
        try{
            Context context = Context.enter();
            context.setOptimizationLevel(-1);
            Scriptable scriptable = context.initStandardObjects();
//            Evaluating the string
            String final_result = context.evaluateString(scriptable, data,
"Javascript", 1, null).toString();
            if (final_result.endsWith(".0")) {
                final_result = final_result.replace(".0", "");
            }
            return final_result;
        } catch (Exception e) {
            return "Calculation error";
        }
    }


}
```